

SOUTHEND, SK, Canada

WMO: 714510

Lat: 56.337N

Lon: 103.287W

Elev: 1129

StdP: 14.11

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.6	-32.6	-44.5	0.4	-37.0	-39.4	0.6	-32.4	14.8	0.2	14.1	-3.5	3.2	240	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	81.5	62.8	78.1	61.2	74.9	60.1	65.8	75.5	64.0	73.3	62.3	70.7	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	88.9	68.8	60.7	82.7	66.8	58.9	77.3	65.2	31.3	75.9	29.8	73.4	28.5	70.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	13.2	11.7	-43.0	89.0	4.6	3.1	-46.3	91.3	-49.0	93.1	-51.5	94.8	-54.8	97.1		
(5)			-43.5	69.4	4.6	2.5	-46.8	71.3	-49.5	72.7	-52.0	74.1	-55.3	76.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	-6.0	-1.7	11.5	29.1	43.0	55.7	61.4	58.8	48.5	34.5	14.4	-1.1
	DBStd	26.45	15.22	14.24	15.12	12.66	9.25	7.23	5.53	6.09	7.85	8.02	12.71	15.49
	HDD50	8552	1736	1446	1195	629	257	26	1	6	121	484	1067	1584
	HDD65	13129	2201	1866	1660	1076	682	288	138	209	497	946	1517	2049
	CDD50	951	0	0	0	4	40	198	355	277	75	2	0	0
	CDD65	55	0	0	0	0	1	10	27	15	2	0	0	0
	CDH74	884	0	0	0	1	46	189	391	224	33	0	0	0
	CDH80	170	0	0	0	0	7	32	77	48	6	0	0	0
	WSAvg	4.7	4.5	4.6	4.9	5.1	5.3	4.7	4.3	4.5	4.7	5.0	4.5	4.2
	PrecAvg	21.9	1.3	0.7	1.1	0.9	1.8	2.7	3.9	3.0	2.4	1.6	1.1	0.9
(15) Precipitation	PrecMax	34.6	7.2	1.7	4.3	1.9	3.8	5.9	8.5	7.7	4.6	3.4	2.7	3.7
	PrecMin	14.7	0.1	0.1	0.0	0.2	0.1	0.6	0.6	0.4	0.5	0.3	0.4	0.3
	PrecStd	5.1	1.5	0.4	1.0	0.5	0.9	1.3	2.0	1.7	1.1	0.8	0.6	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	36.4	51.2	67.7	79.2	83.9	86.9	85.7	78.5	62.9	42.2	32.9
		MCWB	31.7	33.0	40.2	50.5	57.0	63.0	65.8	66.0	61.0	50.6	37.2	31.7
	2%	DB	27.5	31.4	44.5	59.4	72.9	79.1	81.5	80.0	71.6	55.2	36.8	28.3
		MCWB	26.6	29.8	36.7	45.7	53.4	60.1	63.6	63.2	58.2	47.1	34.2	27.4
	5%	DB	22.3	26.3	38.5	53.0	67.3	75.3	78.3	75.7	66.8	49.9	33.5	24.0
		MCWB	21.7	24.9	33.4	41.8	51.1	58.5	61.9	61.7	56.2	45.2	32.2	23.3
	10%	DB	16.4	20.9	33.7	47.6	61.4	71.2	75.0	71.8	61.8	46.0	30.9	20.4
		MCWB	15.8	19.9	30.8	38.9	48.5	57.1	61.3	60.5	54.5	42.8	29.7	19.9
	0.4%	WB	32.3	33.2	41.3	51.1	57.8	67.9	68.9	67.6	63.4	52.9	37.8	31.6
		MCDB	33.3	34.8	50.2	66.6	74.1	77.9	78.9	78.0	74.9	58.4	41.1	32.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.7	29.5	37.4	45.7	54.7	63.4	66.2	65.2	60.0	48.4	34.6	27.4
		MCDB	27.5	30.7	43.5	57.5	67.8	71.8	75.9	74.8	68.7	53.6	36.3	28.1
	5%	WB	21.8	24.4	34.0	42.3	52.1	60.7	64.5	63.4	57.6	45.7	32.3	23.2
		MCDB	22.4	25.7	37.9	51.2	65.0	70.5	74.1	72.4	64.6	49.7	33.7	23.9
	10%	WB	16.2	19.4	30.6	39.4	49.2	58.5	62.8	61.4	55.1	42.9	29.8	19.8
		MCDB	16.7	20.5	33.8	46.7	59.3	68.1	71.3	69.0	61.1	46.1	30.9	20.4
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	22.2	25.5	23.8	22.9	22.4	21.7	20.1	18.3	12.8	13.8	15.7
		MCDBR	20.5	22.9	27.7	30.2	34.9	30.5	29.2	27.7	26.8	21.2	14.0	16.5
		MCWBR	19.4	20.5	20.5	17.7	18.0	14.4	13.6	13.7	15.1	14.3	11.7	15.6
	5% WB	MCDBR	20.0	22.2	24.9	27.4	30.3	24.4	23.8	23.8	23.8	19.9	13.3	16.1
		MCWBR	19.3	20.6	18.3	16.6	16.4	12.5	12.6	13.2	14.3	13.9	11.3	15.7
(40) Clear-Sky Solar Irradiance	taub		0.228	0.247	0.268	0.315	0.327	0.366	0.394	0.356	0.312	0.287	0.237	0.214
	taud		2.255	2.301	2.305	2.352	2.493	2.413	2.356	2.477	2.585	2.563	2.386	2.204
	Ebn at Noon		225	267	288	287	290	278	266	272	272	252	224	198
	Edh at Noon		16	23	30	33	31	34	36	30	23	18	14	12
(44) All-Sky Solar Radiation	RadAvg		239	580	1105	1608	1778	1800	1696	1380	920	463	214	142
	RadStd		24	39	70	107	176	147	113	108	118	61	22	18

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page